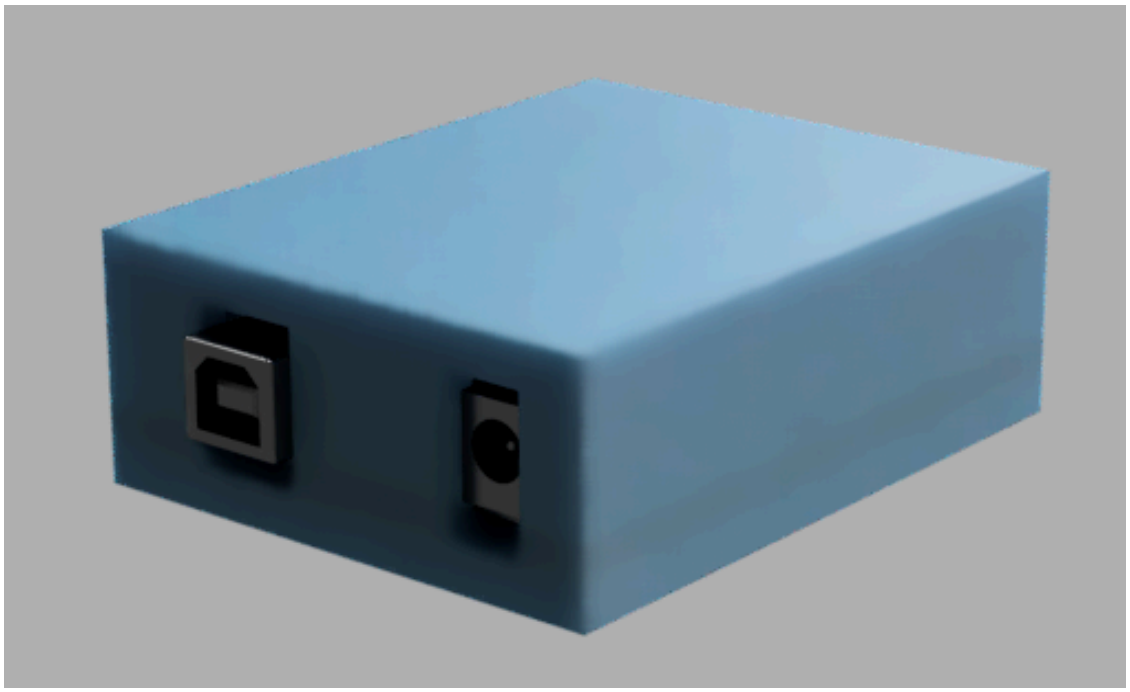
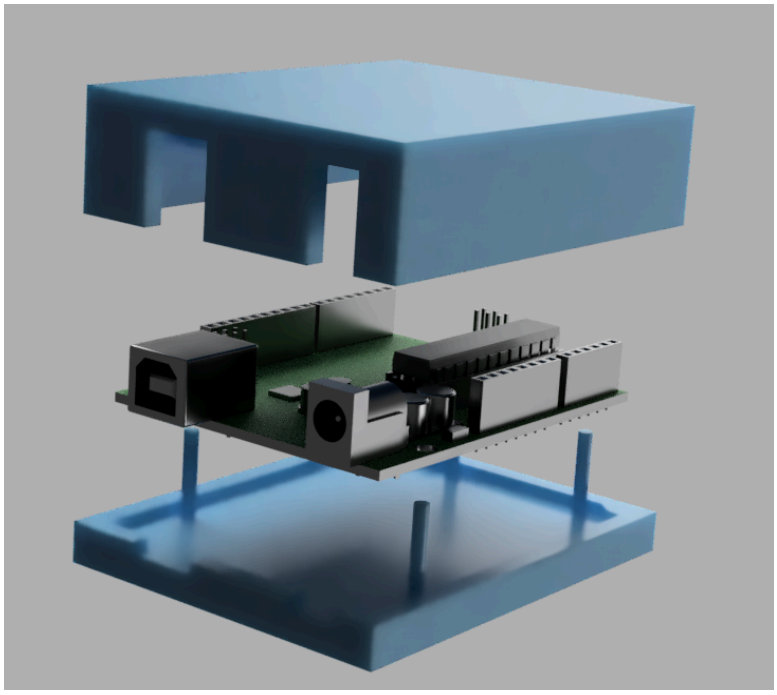
Enclosure for Arduino UNO in Fusion

A portable enclosure for Arduino Uno with internal PCB mounting and 3D-print-ready structure.



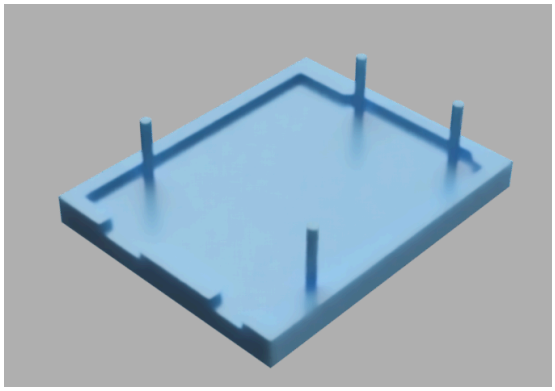


Here from the top to the bottom:

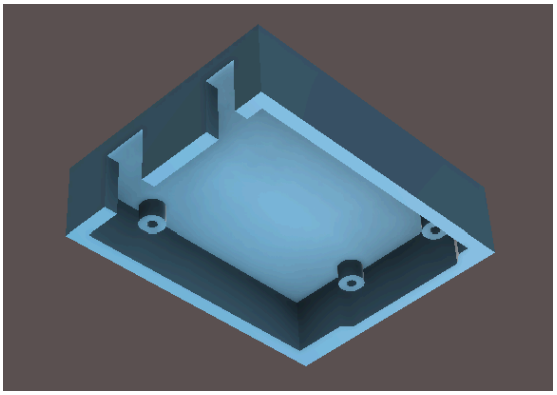
1 - Top of the case

2 - Bottom of the case

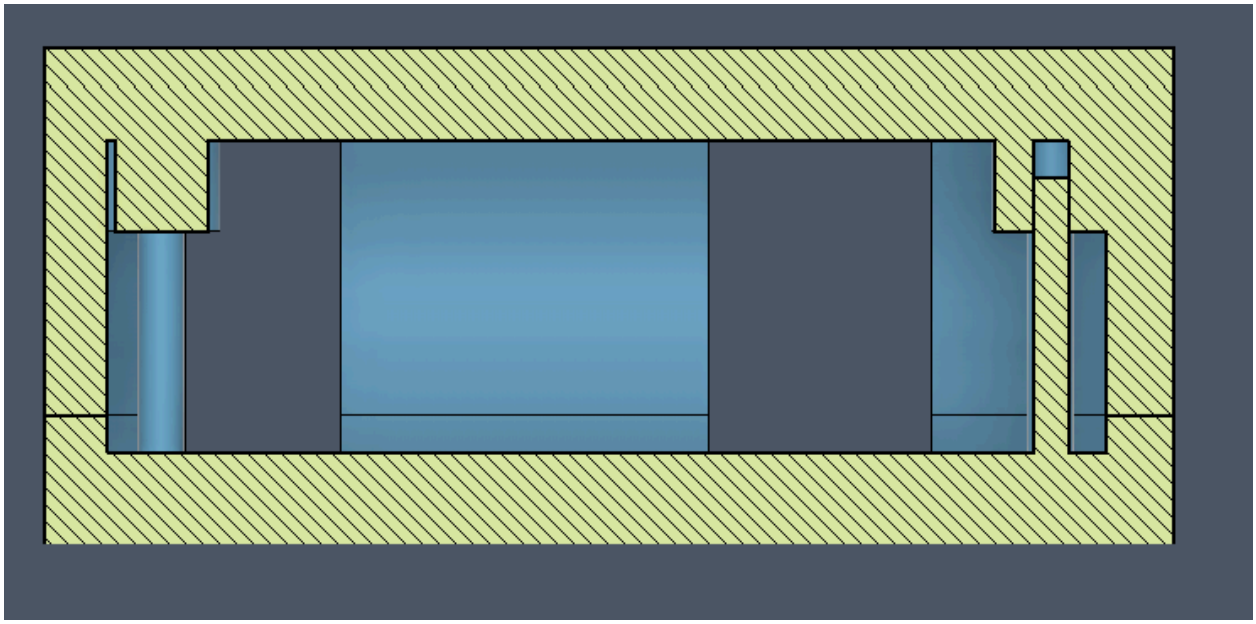
3 - Sketch, representing Arduino Uno



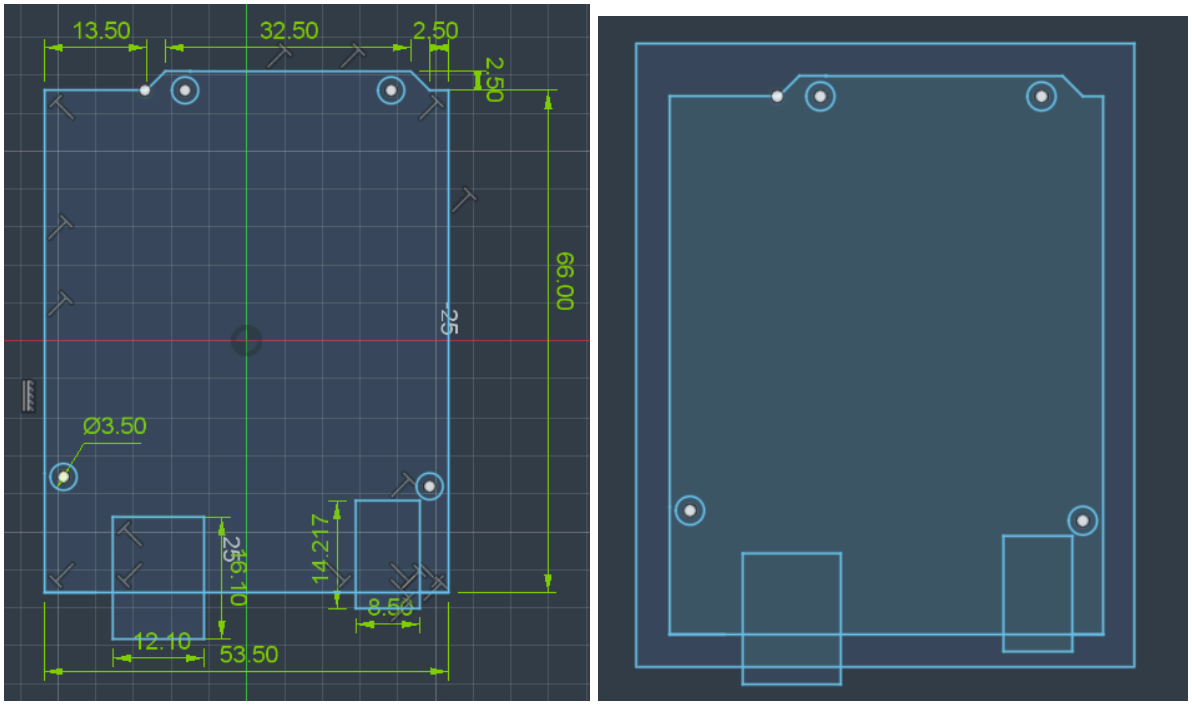
Bottom part of the enclosure with standoffs



Top part with pegs



Analysis of the parts connection



Sketch of the Arduino UNO and frame

Designed a portable enclosure for Arduino Uno to protect the board and allow easy access to USB and Power ports.

The design process began with creating a sketch based on the exact dimensions of the Arduino Uno. A frame extending 4 mm beyond the initial sketch was then added.

Top and bottom parts of the enclosure are designed so they can be connected with standoffs and pegs.

The model was optimized for 3D printing, taking into account manufacturing constraints. The enclosure is designed with a wall thickness of 4 mm and an additional 0.5 mm clearance to compensate for printing tolerances and ensure proper fit.

The enclosure is designed for manufacturing using ABS plastic.